

Sheet 1 - Machine Learning

Q1 : find the global minimum for the following functions

1- $f(x) = 3x^2 + 82x - 9.91$

2- $f(x) = \frac{1}{3}x^3 + 2x^2 + 2x + 3$

Q2 : using Newton's method find the value that make $f(x) = 0$ for the following functions

1 - $f(x) = 3x^2 + 3x - 1$

2 - $f(x) = \frac{1}{3}x^3 + 2x^2 + 2x + 3$

Q3 : using Newton's method find the global minimum for the following functions

1- $f(x) = 3x^2 + 82x - 9.91$

2- $f(x) = 6x^4 + 12x^3 + 24x^2 + 48x + 96$

Q4 : study effect of α, β in equation $f(x) = \alpha + \beta x$

Q5 :

Number of rooms	area	price
4	220	300
3	190	250
3	150	220
2	140	180
3	110	200

A] find X matrix

B] find Y matrix

C] Find correlation matrix between data columns

D] assume initial weights are zeros , what is the MAE at first iteration ?

E] what is values of weights after first update of weights using Gradient Descent (assume learning rate with 0.1)

F] Find the optimal weights using normal equation

Q6 : prove that

$$W_{new} = W_{old} - \frac{\alpha}{2M} \sum_{i=1}^m \frac{h - y}{|h - y|} x_j$$